

# Hallu × Allianz Motor Insurance

## Integration Status Report

Date: 13 July 2026 | Prepared by: Hallu Development Team | Project: Motor Insurance Online Renewal

### Executive Summary

The end-to-end motor insurance renewal flow has been implemented and tested against the **Allianz UAT environment**. The core flow — from vehicle lookup through quotation, customer details, payment (SenangPay sandbox), and Allianz policy submission — is **functional and verified with real customer data**.

**Three items require Allianz's attention before production readiness.**

### 1. Current Flow Status

| Step | Feature                      | Status             | Notes                                                        |
|------|------------------------------|--------------------|--------------------------------------------------------------|
| 1    | Landing Page & Quote Form    | Complete           | NRIC, plate number, postcode, contact, PDPA consent          |
| 2    | Vehicle Lookup + UBB Check 1 | Complete           | Returns vehicle info, NVIC list, NCD. Blocks referral cases. |
| 3    | Variant Selection (MV / AV)  | Complete           | Market Value via NVIC, Agreed Value via LOV.                 |
| 4    | Quotation                    | Complete           | POST /quote. Premium breakdown + add-on covers.              |
| 5    | Add-on & Quote Update        | Complete           | PUT /quote. Windscreen, CART, ERW, driver plans.             |
| 6    | UBB Check 2 (Pre-Quote)      | Blocked            | See Issue #1 — "Invalid Agent Code" from API                 |
| 7    | Customer Details             | Complete           | Policyholder info, address. Auto-derives DOB/gender.         |
| 8    | Payment (SenangPay)          | Complete (Sandbox) | HMAC-SHA256 server-side. Tested and working.                 |
| 9    | Policy Submission            | Complete           | /submission with 3x retry. Returns "status: Success".        |
| 10   | Callback (Post-Issuance)     | Pending            | See Issue #2 — awaiting model clarification                  |
| 11   | Thank-You Page               | Complete (UI)      | Policy summary, free-look info. PDF download pending.        |

### 2. Items Requiring Allianz Input

#### Issue #1: UBB Check 2 — "Invalid Agent Code"

**Endpoint:** POST /v1/openapi/mci/checkUBB

**Symptom:** API returns { "Status": 500, "Error": "Invalid Agent Code" }

**Tested with:** SourceSystem = our Partner ID, and SourceSystem = "PTR" — both fail.

**Impact:** UBB Check 2 performs deeper underwriting validation (reconditioned vehicles, high-risk combos). Without it, edge cases bypass risk screening. UBB Check 1 (in vehicle lookup) still catches major blocks.

**What we need from Allianz:**

- Confirm our Partner ID is enabled for /checkUBB in UAT
- Advise the correct SourceSystem value for CheckUBB
- Provide agent code if separate from Partner ID

## Issue #2: Post-Issuance Callback

**What we've built:** Backend endpoint ready with HMAC-SHA256 signature verification. Extracts contractNumber, policyNumber, status, policyPdf, vehicleLicenseId.

**Pending:** Which delivery model applies to our integration?

| Model | Policy PDF                 | Email to Customer       | Our Action                |
|-------|----------------------------|-------------------------|---------------------------|
| A     | Not sent to us             | Allianz emails directly | Store policy number only  |
| B     | Sent as base64 in callback | Allianz emails directly | Store PDF + policy number |
| C     | Sent as base64 in callback | We email customer       | Store PDF + send email    |

**What we need from Allianz:**

- Which callback delivery model applies?
- Provide callback HMAC secret key
- Confirm callback payload structure
- Provide callback URL registration process

## Issue #3: Production Credentials

Before going live, we need the following:

| Item                         | Purpose                     | Status                          |
|------------------------------|-----------------------------|---------------------------------|
| Production API base URL      | Switch from UAT to live     | Pending                         |
| Production OAuth credentials | Consumer Key + Secret       | Pending                         |
| Production Partner ID        | May differ from UAT         | Pending (confirm if same)       |
| CheckUBB enablement          | Resolve Issue #1            | Pending                         |
| Callback HMAC secret         | Verify webhook authenticity | Pending                         |
| Callback URL registration    | Register our endpoint       | Pending                         |
| Postcode LOV                 | Does Allianz provide this?  | Pending (local fallback in use) |

### 3. Verified API Endpoints (Working in UAT)

Tested and confirmed against Allianz UAT environment:

| Endpoint                       | Method | Purpose                          | Result |
|--------------------------------|--------|----------------------------------|--------|
| /v1/oauth/accesstoken          | POST   | OAuth token (client credentials) | Pass   |
| /v1/openapi/mci/vehicleDetails | POST   | Vehicle lookup + UBB Check 1     | Pass   |
| /v1/openapi/mci/quote          | POST   | Generate quotation               | Pass   |
| /v1/openapi/mci/quote          | PUT    | Update quotation (add-ons)       | Pass   |
| /v1/openapi/mci/submission     | POST   | Submit policy after payment      | Pass   |
| /v1/openapi/mci/lov/avVariant  | GET    | Agreed Value variant list        | Pass   |
| /v1/openapi/mci/checkUBB       | POST   | UBB risk screening (Check 2)     | Fail   |

### 4. Payment Gateway (SenangPay)

| Item                          | Status                                                |
|-------------------------------|-------------------------------------------------------|
| Hash generation (HMAC-SHA256) | Complete — server-side, secret not exposed to browser |
| Hash verification on return   | Complete                                              |
| Sandbox integration           | Tested and working                                    |
| Production switch             | Pending — requires production merchant credentials    |
| Return URL config             | Must be set in SenangPay merchant dashboard           |

### 5. Test Verification Summary

Testing performed against Allianz UAT with real customer data (with consent):

| Field                                      | Value               |
|--------------------------------------------|---------------------|
| Vehicle                                    | PROTON S70 (2024)   |
| Engine                                     | 1477 CC             |
| NCD                                        | 55%                 |
| Cover Type                                 | Comprehensive       |
| Base Premium (Market Value)                | RM 1,401.40         |
| With add-ons (windscreen + special perils) | RM 1,666.00         |
| Submission result                          | "status": "Success" |

### 6. Next Steps

1. **Allianz to resolve** CheckUBB agent code issue (Issue #1)

2. **Allianz to confirm** callback delivery model and provide HMAC secret (Issue #2)
3. **Allianz to provide** production credentials when ready (Issue #3)
4. **Hallu team** to implement callback processing once model is confirmed
5. **Hallu team** to switch SenangPay to production once merchant is verified
6. **Hallu team** to perform production smoke test once all credentials provided

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*This document was prepared for the Allianz partnership meeting. For technical questions, please contact the Hallu development team.*